Lab 1

Introduction to Programming - Lab 1

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Answer the theory questions. Follow the instructions in the Practical section for the programming exercises.

Theory

Question 1: Division operators

Compare `/`, `//` and `%` for integer operands. Use `17` and `5` to explain the quotient and remainder. Then discuss `-17 // 5` and `-17 % 5`, including how Python's floor division differs from simply discarding the fractional part.

Question 2: Truth values of basic scalars

Discuss how Python treats `0`, `-2`, `0.0`, `""` and `"0"` when used as conditions. Why is a non-empty string containing a zero character different from the numeric value zero? Give a small `if` example that demonstrates the distinction.

Practical

Submission: provide one complete Python 3 script for each assigned exercise, together with your test cases. Explain important decisions as code comments. Screenshots are not required.

Scope: use variables, scalar values, strings, arithmetic, conditions, and `for` or `while` loops. Built-in input/output, conversion, `range`, `len`, `abs`, `min`, `max`, and basic string operations are permitted. Do not define functions or classes, import libraries, or use lists, tuples, dictionaries or sets. Algorithmic reasoning is allowed in this Practical section.

Input and output: each numeric value is provided on its own line, in the order stated in the question. A word, command or string occupies one whole line. Print only the requested result, without input prompts. In compact examples, `/` marks a line break; the slash is not part of the actual input or output. Test at least one boundary case and one ordinary case in addition to the example. Multiple valid outputs are accepted only where explicitly stated.

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Question 3: Groups of equal magnet codes

Magnets arrive with orientation code 01 or 10. A group is a maximal consecutive run of the same code. Count the groups by remembering the preceding code. A single magnet forms one group.

Input: Read n, then n orientation strings.

Output: Print the group count.

Bounds: 1 <= n <= 1000.

Example: input `5 / 01 / 01 / 10 / 10 / 01`; output `3`.

Question 4: Covering a rectangular floor

A rectangular floor has length n and width m. Square tiles have side length a, must stay parallel to the floor edges and cannot be cut. Tiles may extend past the floor boundary. Find the fewest tiles needed to cover the floor.

Input: Read n, m and a, in that order.

Output: Print the required tile count.

Bounds: 1 <= n, m, a <= 10^6.

Example: input `7 / 5 / 3`; output `6`.

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